Ye Hong

Department of Human Geography
Lund University
Sölvegatan 10, 223 62 Lund, Sweden

ye.hong@keg.lu.se
+46 73 851 12 00
Homepage, ORCID

RESEARCH AREAS

Human mobility: activity-travel analysis, scheduling, prediction, and modeling/simulation.

Sustainable transport: shared, on-demand, and electrified; inequality, accessibility, and segregation.

Spatial data science: spatio-temporal AI, representation learning, and multimodal data integration.

APPOINTMENT

2026– Lund University, Sweden
Assistant Professor, Fellow in AI, Department of Human Geography

EDUCATION

2020–25 ETH Zurich, Switzerland.
Doctor of Sciences (Dr. sc. ETH Zurich).
Data-Driven Modeling of Multifaceted Individual Mobility Behavior

2018–20 ETH Zurich, Switzerland.
MSc ETH Zurich in Geomatics.
Conserved Quantities in Transport Mode Choices and Mobility Patterns of People

2014–18 Sun Yat-sen University, China.
BSc in GIS and RS.
Hierarchical Community Detection and Functional Area Identification

EXPERIENCE

2025–26 ETH Zurich, Switzerland
Postdoctoral Researcher, Mobility Information Engineering Lab

2025–26 University of Zurich, Switzerland
Postdoctoral Researcher, Urban Analytics

2025 University of California, Santa Barbara, United States
Visiting Researcher, Department of Geography

2019–20 ETH Zurich, Switzerland
Research Assistant, Mobility Information Engineering Lab

2017 Guangdong Academy of Sciences, China
Software Engineer Intern, Guangzhou Institute of Geography

PUBLICATIONS

Journal Articles[1]

2026 **Hong, Y.**, Zhang, Y., Schindler, K., and Raubal, M. "Deep Generative Model for Human Mobility Behavior." *Transportation Research Part C: Emerging Technologies*, 192, 105916. 10.1016/j.trc.2026.105916

2026 Zhang, Y., **Hong, Y.**, Gao, S., and Raubal, M. "Bidirectional yet asymmetric causality between urban systems and traffic dynamics in 30 cities worldwide." *Nature Communications*, 17, 4671. 10.1038/s41467-026-71377-0

2025 **Hong, Y.**, Xin, Y., Dirmeier, S., Perez-Cruz, F., and Raubal, M. "A causal intervention framework for synthesizing mobility data and evaluating predictive neural networks." *Transportation Research Interdisciplinary Perspectives*, 31, 101398. 10.1016/j.trip.2025.101398

2024 Jin, T., Wang, K., Xin, Y., Shi, J., **Hong, Y.**, and Witlox, F. "Is a 15-Minute City Within Reach? Measuring Multimodal Accessibility and Carbon Footprint in 12 Major American Cities." *Land Use Policy*, 142, 107180. 10.1016/j.landusepol.2024.107180

2024 Yao, Y., Zhou, J., Sun, Z., Guan, Q., Guo, Z., Xu, Y., Zhang, J., **Hong, Y.**, Cai, Y., and Wang, R. "Estimating China's poverty reduction efficiency by integrating multi-source geospatial data and deep learning techniques." *Geo-Spatial Information Science*, 27(4), 1000-1016. 10.1080/10095020.2023.2165975.

2023 **Hong, Y.**, Stüdeli, E., and Raubal, M. "Evaluating geospatial context information for travel mode detection." *Journal of Transport Geography*, 113, 103736. 10.1016/j.jtrangeo.2023.103736

2023 **Hong, Y.**, Zhang, Y., Schindler, K., and Raubal, M. "Context-aware multi-head self-attentional neural network model for next location prediction." *Transportation Research Part C: Emerging Technologies*, 156, 104315. 10.1016/j.trc.2023.104315

2023 Martin, H.[#], **Hong, Y.**[#], Wiedemann, N.[#], Bucher, D., and Raubal, M. "Trackintel: An open-source Python library for human mobility analysis." *Computers, Environment and Urban Systems*, 101, 101938. 10.1016/j.compenvurbsys.2023.101938.

2023 **Hong, Y.**, Martin, H., Xin, Y., Bucher, D., Reck, D. J., Axhausen, K. W., and Raubal, M. "Conserved quantities in human mobility: from locations to trips." *Transportation Research Part C: Emerging Technologies*, 146, 103979. 10.1016/j.trc.2022.103979.

2023 Yao, Y.[#], Guo, Z.[#], Dou, C., Jia, M., **Hong, Y.**, Guan, Q., and Luo, P. "Predicting mobile users' next location using the semantically enriched geo-embedding model and the multilayer attention mechanism." *Computers, Environment and Urban Systems*, 104, 102009. 10.1016/j.compenvurbsys.2023.102009.

2023 Wiedemann, N.[#], Martin, H.[#], Suel, E., **Hong, Y.**, and Xin, Y. "Influence of tracking duration on the privacy of individual mobility graphs." *Journal of Location Based Services*, 17(4), 370-388. 10.1080/17489725.2023.2239190.

2022 Guan, Q., Yao, Y., Ma, T., **Hong, Y.**, Bie, Y., and Lyu, J. "Under the Dome: A 3D Urban Texture Model and Its Relationship with Urban Land Surface Temperature." *Annals of the American Association of Geographers*, 112(5), 1369-1389. 10.1080/24694452.2021.1972790

2021 Yao, Y., Wang, J., **Hong, Y.**, Qian, C., Guan, Q., Liang, X., Dai, L. and Zhang, J. "Discovering the homogeneous geographic domain of human perceptions from street view images." *Landscape and Urban Planning*, 212, 104125. 10.1016/j.landurbplan.2021.104125

[1][#] Equal Contribution

2021 Yao, Y.[#], Liu, Y.[#], Guan, Q., **Hong, Y.**, Wang, R., Wang, R., and Liang, X. "Spatiotemporal distribution of human trafficking in China and predicting the locations of missing persons." *Computers, Environment and Urban Systems*, 85, 101567. 10.1016/j.compenvurbsys.2020.101567

2021 Zhang, J., Li, X., Yao, Y., **Hong, Y.**, He, J., Jiang, Z., and Sun, J. "The Traj2Vec model to quantify residents' spatial trajectories and estimate the proportions of urban land-use types." *International Journal of Geographical Information Science*, 35(1), 193-211. 10.1080/13658816.2020.1726923

2020 Yao, Y., Wu, D., **Hong, Y.**, Chen, D., Liang, Z., Guan, Q., Liang, X., and Dai, L. "Analyzing the Effects of Rainfall on Urban Traffic-Congestion Bottlenecks." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 13, 504-512. 10.1109/JSTARS.2020.2966591

2019 Chen, D., Zhang, Y., Yao, Y., **Hong, Y.**, Guan, Q., and Tu, W. "Exploring the spatial differentiation of urbanization on two sides of the Hu Huanyong Line – based on nighttime light data and cellular automata." *Applied Geography*, 112, 102081. 10.1016/j.apgeog.2019.102081

2019 Yao, Y., Liu, P., **Hong, Y.**, Liang, Z., Wang, R., Guan, Q., and Chen, J. "Fine-scale intra- and inter-city commercial store site recommendations using knowledge transfer." *Transactions in GIS*, 23(5), 1029-1047. 10.1111/tgis.12553

2019 **Hong, Y.**, and Yao, Y. "Hierarchical community detection and functional area identification with OSM roads and complex graph theory." *International Journal of Geographical Information Science*, 33(8), 1569-1587. 10.1080/13658816.2019.1584806

2018 He, J., Li, X., Yao, Y., **Hong, Y.**, and Zhang, J. "Mining transition rules of cellular automata for simulating urban expansion by using the deep learning techniques." *International Journal of Geographical Information Science*, 32(10), 2076-2097. 10.1080/13658816.2018.1480783

2018 Yao, Y., **Hong, Y.**, Wu, D., Zhang, Y., and Guan, Q. "Estimating the effects of "community opening" policy on alleviating traffic congestion in large Chinese cities by integrating ant colony optimization and complex network analyses." *Computers, Environment and Urban Systems*, 70, 163-174. 10.1016/j.compenvurbsys.2018.03.005

2018 Yao, Y., Zhang, J., **Hong, Y.**, Liang, H., and He, J. "Mapping fine–scale urban housing prices by fusing remotely sensed imagery and social media data." *Transactions in GIS*, 22(2), 561-581. 10.1111/tgis.12330

2017 Liu, X., He, J., Yao, Y., Zhang, J., Liang, H., Wang, H., and **Hong, Y.** "Classifying urban land use by integrating remote sensing and social medias data." *International Journal of Geographical Information Science*, 31(8), 1675-1696. 10.1080/13658816.2017.1324976

2017 Yao, Y., Liu, X., Li, X., Liu, P., **Hong, Y.**, Zhang, Y., and Mai, K. "Simulating urban land-use changes at a large scale by integrating dynamic land parcel subdivision and vector-based cellular automata." *International Journal of Geographical Information Science*, 31(12), 2452-2479. 10.1080/13658816.2017.1360494

Conference Proceedings

2026 Mühlematter, D. J., Che, L., **Hong, Y.**, Raubal, M., and Wiedemann, N. "UrbanFusion: Stochastic Multimodal Fusion for Contrastive Learning of Robust Spatial Representations." In *Proceedings of the 43rd International Conference on Machine Learning (ICML '26)*. Seoul, South Korea. PMLR 306, 91006–91063. proceedings.mlr.press/v306/muhlematter26a

2025 **Hong, Y.**, and Raubal, M. "Causal inference for interpretable and robust deep learning in

mobility analysis." In *25th Swiss Transport Research Conference (STRC '25)*. Monte Verità, Ascona, Switzerland. STRC. 10.3929/ethz-b-000744423.

2024 **Hong, Y.**, Zhang, Y., and Raubal, M. "Towards realistic individual activity location demand synthesis using deep generative networks." In *24th Swiss Transport Research Conference (STRC '24)*. Monte Verità, Ascona, Switzerland. STRC. 10.3929/ethz-b-000683550.

2023 Wiedemann, N., **Hong, Y.**, and Raubal, M. "Predicting visit frequencies to new places." In *Proceedings of the 12th International Conference on Geographic Information Science (GIScience '23)*, (pp.84:1–84:6). Leeds, UK. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. 10.4230/LIPIcs.GIScience.2023.84.

2022 **Hong, Y.**, Martin, H., and Raubal, M. "How do you go where? improving next location prediction by learning travel mode information using transformers." In *Proceedings of the 30th International Conference on Advances in Geographic Information Systems (SIGSPATIAL '22)*, (pp. 1-10). Seattle, Washington, USA. ACM. 10.1145/3557915.3560996

2022 Martin, H., Wiedemann, N., Suel, E., **Hong, Y.**, and Xin, Y. "Influence of tracking duration on the privacy of individual mobility graphs." In *Proceedings of the 17th International Conference on Location-Based Services (LBS '22)*, (pp.78–88). Munich, Germany. Technical University of Munich. 10.3929/ethz-b-000572753

2021 **Hong, Y.**, Xin, Y., Martin, H., Bucher, D., and Raubal, M. "A clustering-based framework for individual travel behaviour change detection." In *Proceedings of the 11th International Conference on Geographic Information Science - Part II (GIScience '21)*, (pp.4:1–4:15). Virtual. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. 10.4230/LIPIcs.GIScience.2021.II.4

2020 Martin, H., Bucher, D., **Hong, Y.**, Buffat, R., Rupprecht, C., and Raubal, M. "Graph-resnets for short-term traffic forecasts in almost unknown cities." In *Proceedings of the NeurIPS 2019 Competition and Demonstration Track*, (pp.153–163). Vancouver, Canada. PMLR. 10.3929/ethz-b-000437682

Preprints

2024 Dirmeier, S., **Hong, Y.**, and Perez-Cruz, F. "Synthetic location trajectory generation using categorical diffusion models." 10.48550/arXiv.2402.12242

2023 Li, J., Xin, Y., **Hong, Y.**, and Raubal, M. "Interpreting Deep Learning Models for Traffic Forecast: A Case Study of UNet." 10.2139/ssrn.4370154

2023 Timans, A., Wiedemann, N., Kumar, N., **Hong, Y.**, and Raubal, M. "Uncertainty Quantification for Image-based Traffic Prediction across Cities." 10.48550/arXiv.2308.06129

2023 Dirmeier, S., **Hong, Y.**, Xin, Y., and Perez-Cruz, F. "Uncertainty quantification and out-of-distribution detection using surjective normalizing flows." 10.48550/arXiv.2311.00377

TALKS AND PRESENTATIONS

Invited Talks

2026 "Modeling Human Mobility in Context: From Digital Traces to Urban and Transport Applications." Seminar, The Bartlett Centre for Advanced Spatial Analysis (CASA), University College London, London, UK. November 06.

2026 "MobilityGen: Deep Generative Model for Human Mobility Behavior." Multiscale Mobilities Symposium. EPFL, Lausanne, Switzerland. September 11.

Conference Presentations

2025 "Causal inference for interpretable and robust deep learning in mobility analysis." 25th Swiss Transport Research Conference (STRC '25). Monte Verità, Ascona, Switzerland. May 14–16.

2025 "Deep Generative Model for Human Mobility Behavior." Pacific Southwest Region (PSR) University Transportation Center (UTC) Annual Congress (PSR UTC '25). Berkeley, California, USA. March 24–25.

2024 "Towards realistic individual activity location demand synthesis using deep generative networks." 24th Swiss Transport Research Conference (STRC '24). Monte Verità, Ascona, Switzerland. May 15–17.

2022 "How do you go where? Improving next location prediction by learning travel mode information using transformers." 30th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL '22). Seattle, Washington, USA. November 01–04.

2021 "A Clustering-Based Framework for Individual Travel Behaviour Change Detection." 11th International Conference on Geographic Information Science (GIScience '21). Virtual. September 27–30.

2021 "Trackintel: An open-source Python library for human mobility modeling and analysis." GeoPython 2021. Virtual. April 22–23.

Poster Presentations

2026 "Toward Population-Scale Synthetic Mobility: Home-Anchored Generative Models for Activity-Travel Simulation." The Nordic Conference on Computational Social Science (CS2Nordics '26). Copenhagen, Denmark. September 21–22.

GRANTS

2027–29 *MobiData Europe: Unifying Comparative Human Mobility Research*. UZH Global Champion Fund, joint UZH–Lund University partnership grant. CHF 55,795. Role: Co-PI (Lund University).

AWARDS

2026 Global Research Fellow (AI), Lund University Programme for Global Excellence

2020 Culmann-funds (outstanding master thesis), ETH Zurich

2019 Traffic4cast 2nd Place, NeurIPS 2019 challenge

2018 Outstanding bachelor thesis, Sun Yat-sen University

2018 Excellent national undergraduate's creative project, Ministry of Education of China

2017 Excellent student scholarship, Sun Yat-sen University

TEACHING

I have taught university courses at ETH Zurich and Lund University in GIS, spatial data science, mobility analysis, and geoinformation technologies, serving as course leader, lecturer, and head TA. I have

supervised BSc/MSc theses and projects in geoinformation science, mobility modelling, and AI, and delivered invited lectures at the University of California, Santa Barbara.

Lecturer and Head TA

2026 GIS: Geographical Information System for the Social Sciences (Freestanding Master's course, Lund University; 20 students): Course leader and lecturer.

2022–25 Geoinformation Technologies and Analysis (BSc Geospatial Engineering, ETH Zurich; 11–23 students): Lecturer for spatial data science, spatial database, and movement analysis; Head TA; Course and project coordination.

2025 Spatial Data Science (MSc Geomatics, ETH Zurich; 3 students): Lecturer for literature seminar and discussion.

2023–25 Advanced GIS (MSc Geomatics, ETH Zurich; 11–12 students): Lecturer for Geosensors and Volunteered Geographic Information.

2020–21 GIS and Geoinformatics Lab (MSc Geomatics, ETH Zurich; 5–11 students): Head TA.

TA and Course Support

2020–21 GIS III (MSc Geomatics, ETH Zurich; 12–13 students): TA for Geosensors and Volunteered Geographic Information.

2020 Geomatics Seminar (MSc Geomatics, ETH Zurich; 11 students): TA for project supervision.

2020 Project GIS & Cartography (BSc Geospatial Engineering, ETH Zurich; 32 students): TA for course project supervision.

Student Project and Thesis Supervision

2026 Liu, B. "WHOMOLLM: Inferring Sociodemographic Information from Passively Generated GPS Trajectories Using Large Language Models." Master thesis. University of Zurich.

2025 Mühlematter, D. J. "UrbanFusion: Stochastic Multimodal Fusion for Contrastive Learning of Robust Spatial Representations." Master thesis. ETH Zurich.

2025 Kosch, K. "E-Bike Detection for Digital Sports Competitions." Geomatics Master Project. ETH Zurich.

2023 Mühlematter, D. J. "A Reinforcement Learning Environment for Vehicle-to-Grid Charging Strategies in Car-Sharing." Geomatics Master Project. ETH Zurich.

2023 Vuadens, E. "Heuristic Optimization of Smart Charging and Vehicle-to-Grid for an Electric Car-sharing Fleet." Master thesis. ETH Zurich.

2023 Li, J. "Robust Cross-Domain Traffic Prediction Through Invariant Causal Mechanisms." Master thesis. ETH Zurich.

2022 Stüdeli, E. "Evaluating Geospatial Context Information for Transport Mode Detection." Bachelor thesis. ETH Zurich.

2022 Timans, A. "Uncertainty Quantification for Image–based Traffic Prediction." Master thesis. ETH Zurich.

2021 Li, J. "Enhancing the Interpretability of Deep Learning Models for Traffic Forecast Through Visual Analytics." Geomatics Master Interdisciplinary Project. ETH Zurich.

Invited Guest Lectures

2025 "Data-driven modeling of multifaceted individual mobility behavior." GeoG 211C Travel Behavior Analysis (Graduate, Department of Geography, UC Santa Barbara).

2025 "Mobility modeling from individual tracking data." GeoG 111B Transportation Modeling and Simulation (Undergraduate, Department of Geography, UC Santa Barbara).

SERVICE

Conference Organization

Program Committee Member, IEEE International Conference on Big Data (IEEE BigData 2026).

Organizer, 1st ACM SIGSPATIAL International Workshop on Reproducibility in Tracking Data Analysis and Mobility Research, at SIGSPATIAL '23. Hamburg, Germany. Nov 13–16, 2023.

Academic Journal Peer Review

I serve as an ad-hoc reviewer for journals in transport, geoinformation science, urban studies, and artificial intelligence, including:
ACM Transactions on Spatial Algorithms and Systems, *Annals of Regional Science*, *Artificial Intelligence Review*, *BMC Public Health*, *Cities*, *Computers, Environment and Urban Systems*, *Connection Science*, *Data & Policy*, *Ecological Indicators*, *Engineering Applications of Artificial Intelligence*, *Environment and Planning B: Urban Analytics and City Science*, *GIScience & Remote Sensing*, *Humanities and Social Sciences Communications*, *IEEE Access*, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Computational Social Systems*, *IEEE Transactions on Geoscience and Remote Sensing*, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Mobile Computing*, *IEEE Transactions on Vehicular Technology*, *Information Processing and Management*, *International Journal of Digital Earth*, *International Journal of Geographical Information Science*, *International Journal of Health Geographics*, *International Journal of Sustainable Transportation*, *ISPRS International Journal of Geo-Information*, *Journal of Spatial Science*, *Journal of Traffic and Transportation Engineering (English Edition)*, *Journal of Transport Geography*, *Journal of Urban Management*, *Landscape and Urban Planning*, *Nature Communications*, *Neurocomputing*, *npj Urban Sustainability*, *PLOS One*, *Scientific Data*, *Scientific Reports*, *SoftwareX*, *Sustainable Cities and Society*, *Sustainability*, *Transactions in GIS*, *Transportation*, *Transportation Research Interdisciplinary Perspectives*, *Transportation Research Part C: Emerging Technologies*, *Transportation Safety and Environment*, *Transportmetrica A: Transport Science*, *Travel Behaviour and Society*, *Urban Climate*, and *Urban Informatics*.

Updated November 2026